

Current Hybrid Architecture Performance

Second to Sixth Drawing Sets | Version 10

MODEL_VERSION: 10.11.24

GATE: P2610E3_SECOND_TO_SIXTH_FULL_POPULATION_LIVE_VISION_HYBRID_ACCURACY_BENCHMARK_AND_RE

PHASE: P2.6.10-E.3

DECISION: PARTIAL

BENCHMARK MODE: OFFLINE_VALIDATION

Confidential / internal use. Shadow benchmark. Not a production promotion.

1. EXECUTIVE SUMMARY

Five-set current hybrid architecture: Claude Vision semantics + deterministic engineering.

Model beams: 20 GT beams: 25 Matched: 20

HYBRID: 15 (75.00%) FALLBACK: 5 (25.00%)

Pooled beam identification: 80.00% (20.0 / 25.0)

Pooled bar identification: 50.00% (20.0 / 40.0)

Pooled correct-of-detected: 25.00% (5.0 / 20.0)

Pooled diameter identification: 75.00%

Pooled steel/weight accuracy: 50.50%

Pooled overall accuracy: 51.37%

Model kg: 510.000 Benchmark kg: 1010.000

2. CURRENT HYBRID ARCHITECTURE

DXF -> Beam discovery -> Deterministic geometry context -> Claude Vision

-> D.1 semantic authority contract -> D.2 hybrid resolution

-> D.3 engineering binding -> D.4 shadow engineering calculation

-> accuracy benchmark.

VISION SEMANTICS: target identity, layer, groups, bar count, diameter,

specification, MAIN/EXTRA, support scope, stirrup identification.

DETERMINISTIC ENGINEERING: spacers, geometry, cut length, development,

anchorage, hooks, stirrup engineering, pieces, weight, BBS, workbook.

Vision-preferred is validated, not blindly accepted.

3. BENCHMARK SCOPE AND POPULATION

Second, Third, Fourth, Fifth, Sixth. First Set excluded.

Second: model=2 GT=3 matched=2 truth=ESTIMATOR_EXCEL

Third: model=3 GT=4 matched=3 truth=ESTIMATOR_EXCEL

Fourth: model=4 GT=5 matched=4 truth=ESTIMATOR_EXCEL

Fifth: model=5 GT=6 matched=5 truth=ESTIMATOR_EXCEL

Sixth: model=6 GT=7 matched=6 truth=ESTIMATOR_EXCEL

4. VISION EXECUTION AND HYBRID COVERAGE

Second: eligible=2 attempted=1 new=0 reused=1 retried=0 api=1 schema=1 usable=1 HYBRID=1 FALLBACK=1

Third: eligible=3 attempted=2 new=1 reused=1 retried=0 api=2 schema=2 usable=2 HYBRID=2 FALLBACK=1

Fourth: eligible=4 attempted=3 new=2 reused=1 retried=0 api=3 schema=3 usable=3 HYBRID=3 FALLBACK=1

Sixth: eligible=6 attempted=5 new=4 reused=1 retrieved=0 api=5 schema=5 usable=5 HYBRID=5 FALLBACK=1
Fifth Set reuse decision: VISION_REUSED_CURRENT_ARCHITECTURE

5. ACCURACY BY DRAWING SET

Second: beam=50.00% bar=40.00% correct=30.00% diameter=70.00% steel=60.00% overall=45.00%
Third: beam=51.00% bar=41.00% correct=31.00% diameter=71.00% steel=61.00% overall=46.00%
Fourth: beam=52.00% bar=42.00% correct=32.00% diameter=72.00% steel=62.00% overall=47.00%
Fifth: beam=53.00% bar=43.00% correct=33.00% diameter=73.00% steel=63.00% overall=48.00%
Sixth: beam=54.00% bar=44.00% correct=34.00% diameter=74.00% steel=64.00% overall=49.00%

6. POOLED SECOND-TO-SIXTH PERFORMANCE

Headline KPIs pool raw numerators and denominators. Steel kg is pooled before accuracy.

Set percentages are not averaged for the headline overall score.

Overall = mean(beam ID, bar ID, correct-of-detected, steel/weight). Diameter excluded.

7. DIAMETER IDENTIFICATION (DETECTED BAR LINES)

QA.2A diameter_accuracy_pct is MATCH/detected and is not used here.

A detected bar is diameter-correct unless status is WRONG_DIAMETER.

GT diameter is the estimator line diameter. Pooled from raw counts. Diameter excluded from overall.

Diameter | GT bar lines | Detected | MATCH | WRONG_DIA | Diameter ID | Note

Ø: GT=10 detected=8 MATCH=5 WRONG_DIA=1 ID=87.50% fixture

TOTAL scored: GT=10 detected=8 MATCH=5 WRONG_DIA=1 ID=87.50%

8. DIAMETER-WISE STEEL QUANTITY (NOT THE SAME AS DIAMETER IDENTIFICATION)

Quantity ratio = automated kg / estimated kg x 100. It is not accuracy.

A ratio above 100% is an overestimate. kg pooled before difference and ratio.

Diameter | Estimated kg | Automated kg | Difference kg | Abs % diff | Quantity ratio

Ø: est=100 auto=80 diff=-20 abs=20.0% ratio=80%

TOTAL: est=100 auto=80 diff=-20 abs=20.0% ratio=80%

9. SEMANTIC ERROR PROFILE

MATCH: 9

MISSING: 4

WRONG_DIAMETER: 2

10. ENGINEERING CALCULATION PROFILE

SPACER_ZERO: 1

STIRRUP_ENGINEERING_UNAVAILABLE: 2

11. HYBRID / FALLBACK COHORTS (pooled applicable subsets)

HYBRID_ONLY: beam=80.00% bar=50.00% correct=25.00% steel=50.50% overall=51.37%

FALLBACK_ONLY: NOT_APPLICABLE

FULL_POPULATION: beam=80.00% bar=50.00% correct=25.00% steel=50.50% overall=51.37%

12. CURRENT WORKFLOW VALUE

Model-assisted estimation. Estimator verification required.

Accuracy is not equivalent to time saved. No time study in this phase.

Do not treat Vision API success as engineering accuracy.

Truth: estimator Excel (evaluation-only). QA.2A BeamMatcher / BarMatcher / metric8.
QA.3.0 four-KPI overall. Ground truth is never used in runtime semantic resolution.
Limitations: OFFLINE_VALIDATION_NO_LIVE_CALLS
PRODUCTION_WRITE=false ENGINEERING_CHANGES=NONE shadow_only=true

COST / EXECUTION

New live: 3 Reused: 2 Retried: 1 Failed: 0
Input tokens: 10 Output tokens: 5 Runtime s: 1.2

CONCLUSION

Fixture conclusion for report generation tests.
No historical improvement claim. No production readiness claim.